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al health care network to ensure adequate dental care services [1].

Within the scope of the Unified Health System, immediate relief of the user's suffering through root access is the responsibility of dentists and Brazilian Oral Health Teams (eSB) who work in Primary Health Care (PHC) or emergency care services. Subsequently, the referral to the CEO [2] is performed, so that the treatment can be completed in specialized endodontic care. However, due to difficulty in access or care discontinuity, some treatments are not completed [3], compromising their resolution.

Studies indicate that municipalities with a consolidated Oral Health Care Network have better performance in the provision of dental care [4-7]. However, most studies [3,8-12] that evaluated the performance of specialized endodontic services in dental care centers focused on one or two components or elements restricted to specialized care, without considering their relationship with other services in the care network. In the logic of health care networks, case resolution can be assessed by the service's ability to meet demand and refer cases that require more specialized care [13,14].

Given the complexity of explaining the factors that influence the resolution of health services, it is necessary to conduct analyses that integrate the levels of care required for oral health care continuity. In this context, using a wide range of indicators, this study aimed to identify factors associated with the resolution of endodontic treatment in Brazilian municipalities with CEO. The hypothesis to be examined in this study is that municipalities in a more favorable social context and with better health system characteristics achieve better completion rates for endodontic treatments.

■ Material and Methods

Study Design

This is an ecological study at the municipal aggregation level, with analysis of secondary data from SUS information systems [15] and the external evaluation of the second cycle of the National Program for Improving Access and Quality in Specialized Dental Centers (PMAQ-CEO) [16].

Data Collection

Cities with at least one CEO implemented during the period of the external evaluation of the second cycle of PMAQ-CEO that had joined the aforementioned program and that had complete data regarding Modules I, II, and III of the external evaluation instrument [16] were included. Municipalities that joined the second cycle of PMAQ-CEO but did not participate in the external evaluation, and those that did not receive certification from the Ministry of Health, were excluded.

The study has the resolution of endodontic treatments in SUS as the dependent variable, measured by the ratio of completed endodontic treatments to endodontic access procedures performed by the dentist. To calculate this ratio, the sum of the following endodontic procedures was considered as the numerator, according to the SIA/SUS database: filling of deciduous teeth (0307020037); filling of permanent teeth with two roots (0307020045); filling of permanent teeth with three or more roots (0307020053); filling of permanent teeth with one root (0307020061); endodontic retreatment in a permanent tooth with two roots (0307020088); endodontic retreatment in a permanent tooth with three or more roots (0307020096); and endodontic retreatment in a

Table 1. Dependent variable, categorization, and source for obtaining data.

Dependent Variable	Calculation Formula	
	Numerator	Denominator
Resolution of endodontic treatments in SUS	Number of completed endodontic treatments performed by the dentist in a given location and period. Source: SIA/SUS	The number of endodontic access procedures performed by the dentist in the same location and period. Source: SIA and SISAB/SUS

Contextual factors and health system characteristics were considered as independent variables. The contextual factors included were: national macroregion (Northeastern; Northern; Mid-Western; Southern and Southeastern); population size of the municipality (less than or equal to 50 thousand inhabitants; between 50,001 and 100 thousand inhabitants; and more than 100 thousand inhabitants) [17]; municipal human development index (HDI-M) (quartiles – up to 0.641; between 0.642 and 0.722; between 0.723 and 0.761; and equal to or greater than 0.762) [17] (Table 2).

Table 2. Independent variables by dimension, categorization, and data source.

Dimension / Indicators	Categories	Source
Contextual		
HDI-M	Up to 0.641	IBGE
	0.642 to 0.722	
	0.723 to 0.761	
	0.762 and more	
Population Size (inhabitants)	Up to 50,000	IBGE
	from 50,001 to 100,000	
	100,001 and more	
Geographic Region	Northern	IBGE
	Northeastern	
	Southeastern	
	Southern	
	Mid-Western	
Health System Characteristics		
Dental plan coverage (%)	Less than 2%	ANS
	2 to 9%	
	10% and more	
Average Oral Health Team Coverage (%)	Continuous variable	e-manager
Number of Dentists linked to SUS/100,000 inhabitants	Less than 30	CNES and IBGE
	30 to 49.9	
	50 and more	
First Dental Consultation Coverage	Up to 8.3%	SIA-SUS and IBGE
	8.3% and more	
Urgent Care Center	Yes/No	CNES
Number of CEO	1	CNES
	2 or more	
PMAQ - CEO Certification	Low financial incentive	PMAQ CEO
	High financial incentive	

IBGE: Brazilian Institute of Geography and Statistics; ANS: National Supplementary Health Agency; CNES: National Registry of Health Establishments; SIA-SUS: Outpatient Information System.

The Health System characteristics considered were: dental plan coverage (less than 2%; 2 to 9% and 10% and more coverage); oral health team coverage; proportion of dentists linked to SUS per 100,000 inhabitants

(less than two; equal to or more than two); PMAQ-CEO classification: performance ing to the second cycle of PMAQ-CEO certification by the Ministry of Health (low financial incentive; high financial incentive) [16] (Table 2).

This last variable, derived from the PMAQ-CEO certification, was generated through the financial resulting in the following classification: disqualified; poor; regular; good; very good; and excellent. In municipalities with more than one CEO, certification was obtained through the classification method in each municipality. For data analysis, this variable was dichotomized into low financial incentive (disqualified/poor/regular) and high financial incentive (good/very good/excellent) – (Table 2).

As a data source, in addition to the external evaluation of the second cycle of PMAQ-CEO [20], the study used information from the Brazilian Institute of Geography and Statistics (IBGE) [21], data from e-manager and the National Registry of Health Establishments (CNES), data from the Brazilian National Supplementary Health Agency (ANS), indicators monitored and evaluated through the SIA/SUS database [15], a system available for feeding and monitoring the outpatient production of municipalities, composing a set of simple and composite indicators that seek to make a contextualized assessment of the SUS performance based on its principles and guidelines according to the proposed study model.

The data used refer to 2018, when the external evaluation of the second cycle of PMAQ-CEO was available.

Data Analysis

Microsoft Excel (Microsoft Corp., Redmond, WA, USA) software was used to organize the database, and the analysis was performed using STATA (StataCorp LLC, College Station, TX, USA), version 13.1. For descriptive analysis, simple and relative frequencies were calculated. To analyze the factors associated with the resolution of endodontic treatment in SUS in Brazilian municipalities, the outcome was categorized into quartiles, and ordinal logistic regression (proportional odds model) was performed, with statistical significance assessed using the Wald test.

The adjusted analysis used a hierarchical model with two levels: distal (contextual factors) and proximal (health system characteristics). At each hierarchical level, "backward" regression was performed, with all level variables entered, and those with $p > 0.20$ were removed from the model. For all analyses, 5% significance level was considered.

■ Results

Of the 5,570 Brazilian municipalities, 914 (16.4%) implemented a CEO during the second cycle of PMAQ-CEO, and of these, 859 (94%) met the inclusion criteria. Excluding those that exceeded the value of 1 in the outcome variable ($n=124$; 14.4%) and those that did not receive certification from the Ministry of Health ($n=51$; 5.9%), the sample consisted of 684 municipalities.

municipalities have less than 10% dental plan coverage (62.6%), and the median oral health Brazilian Family Health Strategy (FHS) was 59.6%. Most municipalities (86.6%) do not have an Urgent Care Center. 23.1% of municipalities have 50 or more dental surgeons in SUS per 100,000 inhabitants, and 39% of municipalities achieved first dental consultation coverage in SUS of 8.3% or more. A higher frequency of municipalities with a CEO (89%) and a higher financial incentive in the assessment of access and quality, according to PMAQ-CEO certification (59.2%), was identified (Table 3).

Table 3. Sample description, according to contextual characteristics and oral health services in Brazilian municipalities.

Variables	N	%
Region		
Northern	42	6.1
Northeastern	238	34.8
Southeastern	261	38.2
Southern	100	14.6
Mid-Western	43	6.3
Population Size		
Up to 50,000	296	43.3
from 50,001 to 100,000	165	24.1
100,001 and more	223	32.6
HDI ^a (quartiles)		
Up to 0.641	172	25.2
0.642 to 0.722	171	25
0.723 to 0.761	174	25.4
0.762 and more	167	24.4
Dental Plan Coverage		
Less than 2%	209	30.6
2 to 9%	219	32
10% and more	256	37.4
Average Oral Health Team Coverage	--	58.0
Dentists per 100,000 Inhabitants		
Less than 30	226	33
30 to 49.9	300	43.9
50 and more	158	23.1
First Dental Consultation Coverage		
Up to 8.3%	417	61
8.3% and more	267	39
Urgent Care Center		
No	592	86.6
Yes	92	13.5
Number of CEO ^b		
1	609	89
2 or more	75	11
PMAQ CEO ^c Classification (Mode)		
Low financial incentive	279	40.8
High financial incentive	405	59.2

^aHDI-M: Municipal Human Development Index; ^bBrazilian Dental Specialty Centers; ^cPMAQ-CEO: National Program for Improving Access and Quality of Specialty Dental Centers.

The median ratio of completed endodontic treatments to endodontic access procedures performed in SUS municipalities was 0.33; that is, for every 10 teeth opened for pain relief due to deep dental caries, approximately three teeth had their endodontic treatment completed (Figure 1).

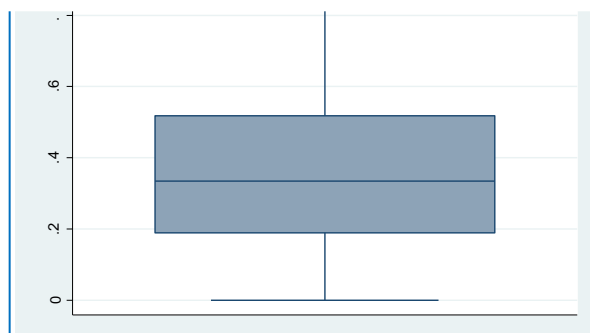


Figure 1. Box plot of variable resolution of endodontic treatment in SUS.

In the crude and adjusted analyses, the Northeastern region stood out, with nearly twice the likelihood of greater resolution than the Southern region. With increases in population size and HDI-M, the resolution of endodontic treatments decreased. In the adjusted analysis, medium-sized municipalities had twice the chance as larger municipalities, and municipalities in the second quartile of HDI-M had 2.5 times the chance as those in the highest HDI quartile (Table 4).

The resolution rate of endodontic treatments was significantly higher in municipalities with lower dental plan coverage, with municipalities with coverage below 2% having a 2.4 times greater chance than municipalities with coverage equal to or greater than 10%. Regarding oral health coverage in the FHS, although municipalities with greater coverage had a higher resolution rate in the crude analysis, this association did not persist in the adjusted analysis. The same occurred with the existence of the Urgent Care Center, a greater number of dentists per 100,000 inhabitants, and greater first dental consultation coverage. Municipalities classified as very good or excellent in the PMAQ-CEO certification had a 40% higher resolution rate than those with worse classifications (Table 4).

Table 4. Crude and adjusted analysis of the resolution of endodontic treatments in SUS according to contextual variables and characteristics of oral health services in Brazilian municipalities.

Level	Variables	Crude Analysis		Adjusted Analysis	
		OR (95%CI)	p-value	OR (95%CI)	p-value
1	Region		< 0.001		0.016
	Southern	1		1	
	Southeastern	0.9 (0.6-1.3)		0.8 (0.5-1.2)	
	Northern	1.6 (0.8-3.1)		1.3 (0.6-2.7)	
	Mid-Western	1.2 (0.7-2.3)		0.9 (0.5-1.7)	
	Northeastern	2.7 (1.8-4.1)		1.9 (1.1-3.4)	
	Population Size		< 0.001**		0.001
	100.001 and more	1		1	
	from 50.001 to 100.000	1.4 (1.0-2.1)		2.0 (1.4-2.9)	
	Up to 50.000	2.6 (1.9-3.6)		1.2 (0.8-1.8)	
	HDI ^a (Quartiles)		< 0.001**		
	0.762 and more	1		1	< 0.001
	0.723 to 0.761	1.6 (1.1-2.4)		1.4 (0.9-2.1)	
	0.642 to 0.722	4.1 (2.8-6.1)		2.5 (1.6-4.0)	
	Up to 0.641	3.8 (2.6-5.7)		1.3 (0.7-2.5)	
2	Dental Plan Coverage		< 0.001		0.003
	10% and more	1		1	
	2 to 9%	1.9 (1.4-2.6)		1.5 (1.0-2.2)	
	Less than 2%	3.9 (2.7-5.4)		2.4 (1.5-4.0)	

more	2.3 (1.5-3.3)	1.2 (0.7-1.9)	
tal Consultation Coverage		< 0.001	0.057
8.3%	1	1	
8.3% and more	1.47 (1.1-1.9)	0.72 (0.5-1.0)	
Urgent Care Center		0.002	0.262
No	1	1	
Yes	0.8 (0.7-0.8)	0.8 (0.5-1.2)	
Number of CEO ^b		0.187	0.219
1	1	1	
2 or more	0.74 (0.5-1.2)	1.4 (0.8-2.4)	
PMAQ CEO ^c Classification (Mode)		0.058	0.003
Disqualified/poor/regular	1	1	
Good/very good/excellent	1.4 (1.0-1.8)	1.4 (1.1-1.9)	

^aHDI-M: Municipal Human Development Index; ^bBrazilian Dental Specialty Centers; ^cPMAQ-CEO: National Program for Improving Access and Quality of Specialty Dental Centers; All variables were included in the adjusted model, except for the number of CEO and the proportion of tooth extractions, as they presented $p > 0.20$ in the crude analysis.

■ Discussion

The results of this study indicate that greater resolution of endodontic treatments in SUS was associated with municipalities in the Northeastern region, with medium-sized populations, intermediate HDI-M, lower dental plan coverage, and better performance in the second cycle of PMAQ-CEO.

Despite the progress in the implementation and provision of specialized care in Brazil, notably in endodontics [3,11,12], the present study shows that in half of the Brazilian municipalities, for every ten open teeth, at least seven do not have endodontic treatment resolved in SUS. This demonstrates that the resolution of this treatment in the Brazilian public system remains low, reflecting lower access to dental specialties than to Primary Health Care.

However, in Brazil, the increase in oral health team coverage in FHS since the 2000s has not been accompanied by the same rate of expansion in access to specialized care, compromising the principle of SUS comprehensiveness and, consequently, the resolution of treatment [22-26].

Since the implementation of the Brazilian Oral Health Policy (PNSB), studies [3,11,12,24-26,28] have evaluated the performance of specialized care in the SUS using various criteria. Regarding performance by region, there is divergence in results. A study analyzing the fulfillment of CEO goals found better performance in the Southeastern region and worse performance in the Southern and Northeastern regions [24]. In another study, based on the average number of procedures, the Southeastern, Northeastern, and Southern regions had a higher proportion of CEO with poor/regular evaluation [11]. In the present study, the Northeastern region presented the most satisfactory result. An ecological study [27] evaluated structural indicators in the Health Units that participated in the second and third cycles of the Primary Care Access and Quality Improvement Program (PMAQ-AB), indicating that the Northeastern region had the most participating teams and the lowest percentage of health units with worsening indicators. The authors suggest that the region's positive results may have been influenced by investments in primary care and by the region's historical preference for the family health care model. It is noteworthy that each study evaluated health services using different measures, making comparisons of the findings difficult. To our knowledge, this is the only study to evaluate oral health care using an indicator that integrates two levels of care: Specialized Dental Care and Primary Health Care.

an integrated manner with Primary Health Care, there is less treatment continuity in these smaller municipalities with lower HDI-M scores than in more socially developed locations with larger populations [3]. These findings demonstrate the relevance of the Brazilian Oral Health Policy (PNSB), especially in smaller and less developed municipalities, and evidence its effect on the expansion and qualification of secondary care, particularly in vulnerable municipalities [28]. The greater resolution of endodontic treatment in SUS, linked to the higher PMAQ-CEO certification score, also highlights the importance of this program in reducing inequities and improving team performance. These results are very relevant, considering their potential to support policies aimed at improving the treatment of dental caries and their significance in achieving more equitable conditions for the population using public oral health services in Brazilian municipalities. Other studies have also demonstrated the program's effect on Primary Health Care, showing greater improvement in the average performance of PMAQ-AB in poorer municipalities across the three cycles, as well as an association between better structure and work processes in public oral health services and better performance [27,29].

Given the nature of the data source used to measure the study outcome, which includes only outpatient procedures performed in SUS, its results only allow for the evaluation of treatments completed in the public system, and it is not possible to include treatments that were initiated in SUS but were completed in the private system and vice versa. This helps explain the association between greater resolution of endodontic treatments in SUS and lower dental plan coverage. It could be inferred that municipalities with greater dental plan coverage may have a higher number of treatments completed outside the SUS, thereby reducing the ratio of completed endodontic treatments to endodontic access procedures performed in the SUS.

In summary, the study's initial hypothesis was partially confirmed, showing greater resolution of endodontic treatments in SUS in municipalities with an intermediate social context but favorable health system characteristics, as demonstrated by superior performance in the PMAQ-CEO outcomes. Given the persistent challenges in the provision and effectiveness of endodontic treatments within Brazil's Unified Health System (SUS), the literature highlights strategic pathways to improve service quality and expand resolvability. Continuous professional development of primary care oral health teams is essential, enabling them to perform low-complexity endodontic procedures with appropriate technologies and well-defined clinical protocols [2,30]. At the same time, the consolidation of integrated care flows between primary care and CEO must be supported by regulation mechanisms based on clinical criteria to ensure the rational use of specialized services [30,31]. The incorporation of specific performance indicators – such as treatment completion rates, waiting times, and retreatment rates – should guide the monitoring and evaluation processes, providing evidence for decision-making by managers and policymakers [31,32]. In this context, increasing the resolvability of endodontic treatments in the SUS requires not only broader access but also strengthened management, clinical quality, and analytical capacity within oral health services [30].

The study's limitations are inherent to its nature: it is an ecological study based on secondary data, which is subject to inaccuracies and limitations in the quality and availability of the information needed to better explain the phenomenon under study. In addition, the causal interpretation of the relationships between variables should be avoided. Another point worth highlighting is that compliance with the inclusion criteria may have excluded municipalities with less-organized health systems (as evidenced by non-adherence to the PMAQ-CEO

dy.

■ Conclusion

Factors associated with a greater chance of resolution were municipalities in the Northeastern region, of medium size, with medium HDI-M, lower dental plan coverage, and better performance in the PMAQ-CEO. Municipalities of all sizes demonstrate greater fragility in care continuity, as reflected in a low proportion of completed endodontic treatments within the Brazilian Unified Health System. These results indicate the need for future research to investigate the underlying reasons for this finding.

■ Authors' Contributions

■ Financial Support

None.

■ Conflict of Interest

The authors declare no conflicts of interest.

■ Data Availability

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